Multivariate Time Series Anomaly Detection and GAVEL

Learning Temporal, Relational, and Global Patterns in Multivariate Time Series for Accurate Anomaly Detection

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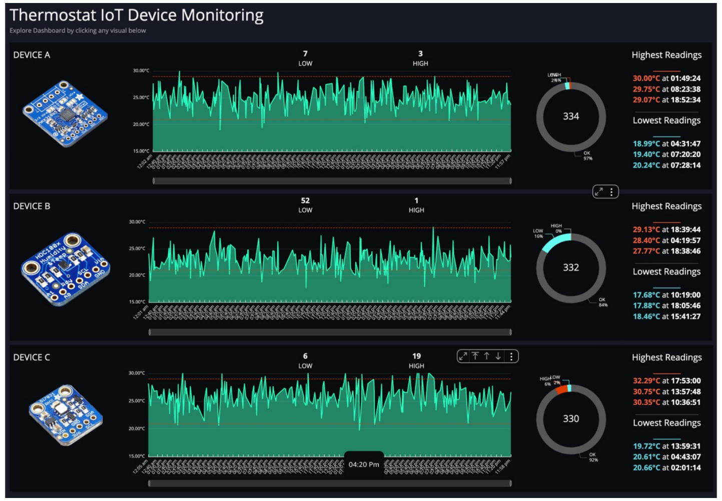
Outline

- Introducing Time-Series Data
- Multivariate Time-Series Anomaly Detection
- Evaluation Metrics and Threshold Selection
- Previous Work
- Motivation and Key Observations
- GAVEL
- Research Questions and Results
- Conclusion

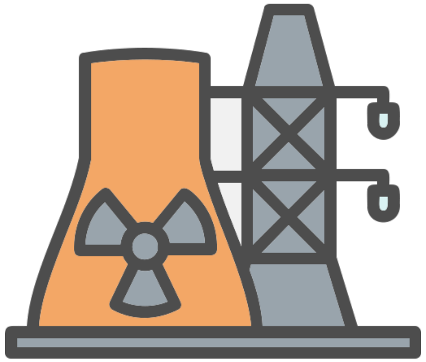
Introducing Time-Series Data

- **Time-Series Data**
  - Data observed sequentially over time
  - Each time point is not independent, but reflects the influence of past observations
  - When multiple series are observed together on the same time axis, the data become multivariate time series
  - Examples
    - Industrial sensor measurements
    - Server / network monitoring logs
    - Physiological signals
    - Financial transaction data



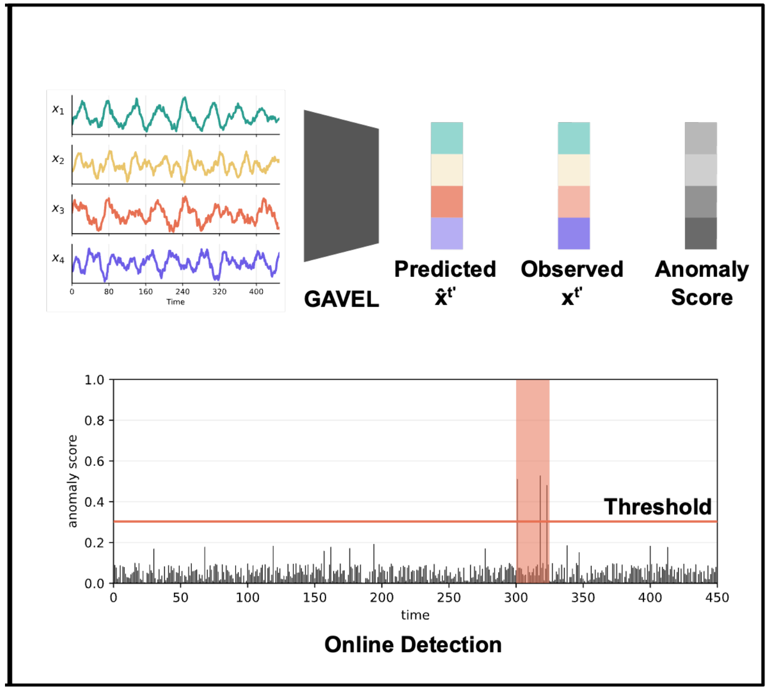
Introducing Time-Series Data

- **Downstream Tasks for Time-Series Data**
  - Forecasting
    - Weather forecasting
    - Power consumption forecasting
    - Traffic forecasting
  - Classification
    - Human activity recognition using smartwatches
  - Anomaly Detection
    - Fault detection in industrial systems
    - Network failure detection
    - Detection of dangerous health conditions



Multivariate Time-Series Anomaly Detection

- **Problem Definition**
  - The goal is to determine whether a specific time point or segment is normal or anomalous
  - To assess whether time step t is anomalous, a window containing t and its preceding observations is used as input
  - The window is slid along the time axis, and anomaly decisions are made for all time steps
- **Why is the Problem Difficult?**
  - Labels are scarce
  - The definition of anomaly is often ambiguous or domain dependent
  - Real-world systems are usually stable, so anomalous events are rare
  - Reproducing anomalies manually can be dangerous or even infeasible
- **Common Approach**
  - Learn normal patterns
  - At inference time, predict what should be observed if the system is normal
  - Actual observation vs Normal prediction as an anomaly score
  - Classify an input as anomalous when the anomaly score exceeds a threshold

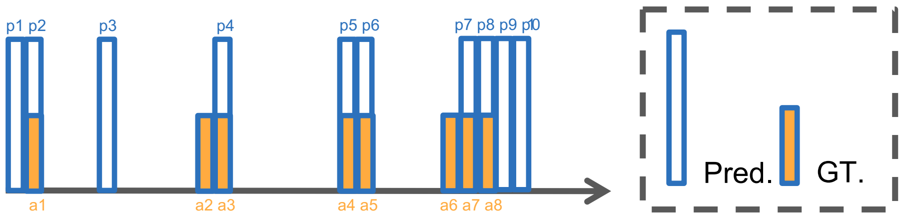


Multivariate Time-Series Anomaly Detection

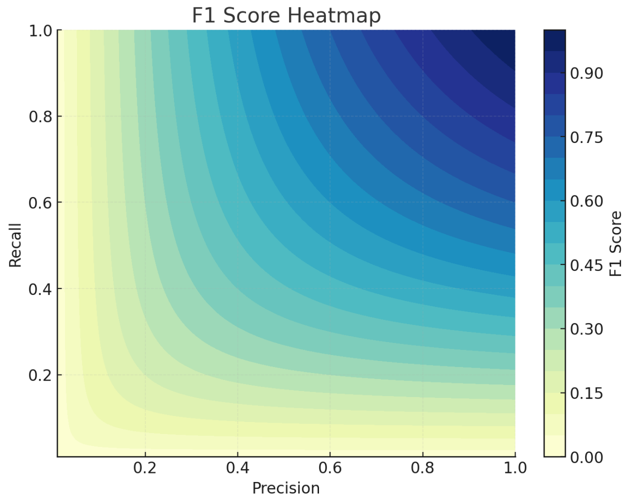
- **How Should We Learn Representations?**
  - From what perspective should the input window be represented?
  - Temporal patterns
  - Inter-series relationships
- **How Should We Define Anomaly Scores and Thresholds?**
  - Trade-off between detection and precise prediction
- **How Should We Evaluate Performance?**
  - What characteristics of the problem does each evaluation metric reflect?

Multivariate Time-Series Anomaly Detection-Evaluation Metrics

- **Precision**
  - $\frac{True\ Positives}{True\ Positives+False\ Positives} = \frac{Correctly\ Detected\ Anomalies}{Predicted\ Anomalies}$
- **Recall**
  - $\frac{True\ Positives}{True\ Positives+False\ Negatives} = \frac{Correctly\ Detected\ Anomalies}{Actual\ Anomalies}$
- **F1 Score**
  - $2 \times \frac{Precision \times Recall}{Precision+Recall}$
  - Balances precision and recall

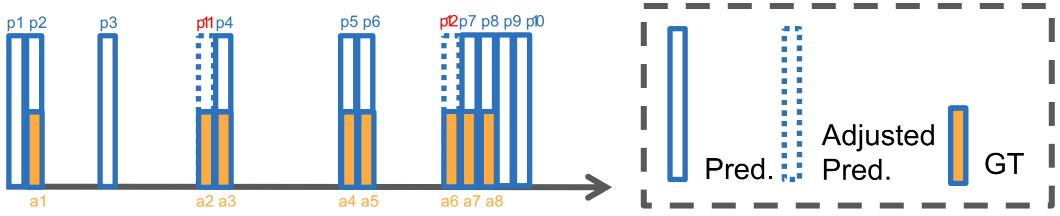


Precision = 0.6, Recall = 0.75, F1 = 0.667



Multivariate Time-Series Anomaly Detection-Evaluation Metrics

- **Point Adjust Precision & Recall**
  - Consecutive anomalies are treated as a single anomaly segment
  - Detecting any point within the segment counts as detecting the entire segment
- **Point Adjust Precision**
  - $\frac{\mathit{Correctly\ Detected\ Anomalies}}{\mathit{Predicted\ Anomalies}}$ (Predictions = Predictions + Adjusted Predictions)
- **Point Adjust Recall**
  - $\frac{\mathit{Correctly\ Detected\ Anomalies}}{\mathit{Actual\ Anomalies}}$



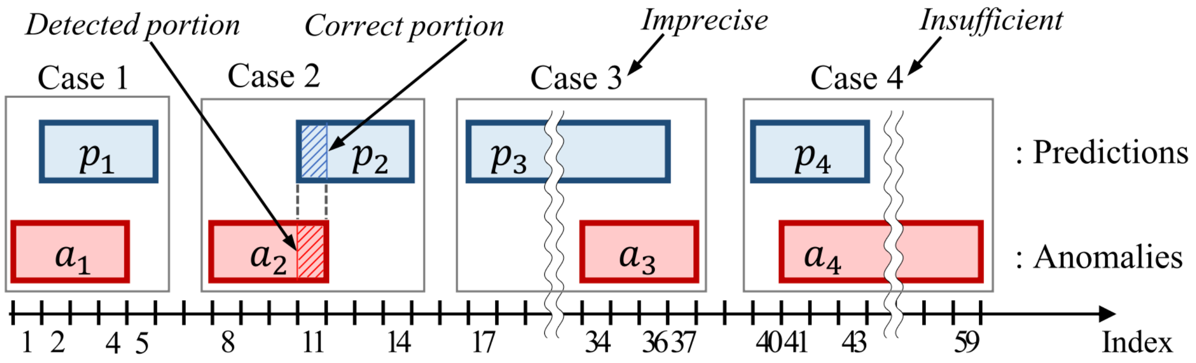
PA-Precision = 0.75, PA-Recall = 1
F1 = 0.857

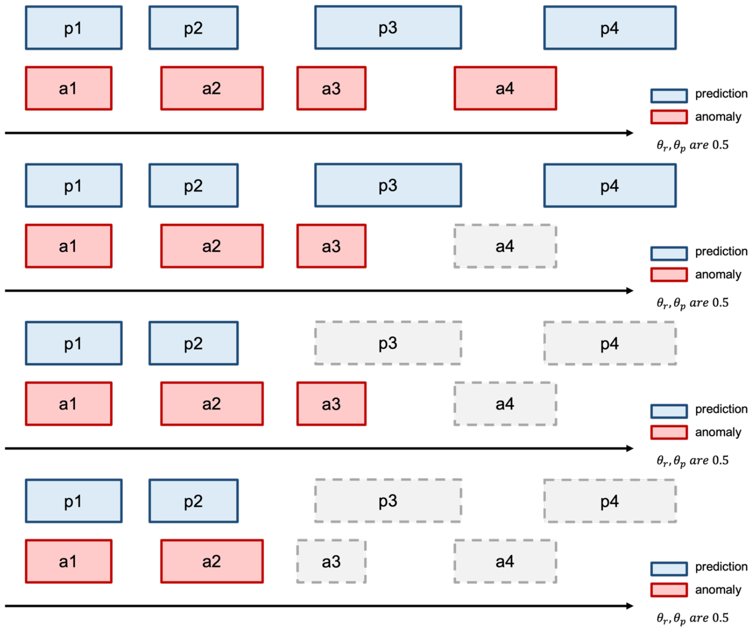
truth	0	0	1	1	1	0	0	1	1	1
score	0.6	0.4	0.3	0.7	0.6	0.5	0.2	0.3	0.4	0.3
point-wise alert	1	0	0	1	1	1	0	0	0	0
adjusted alert	1	0	1	1	1	1	0	0	0	0

Multivariate Time-Series Anomaly Detection-Evaluation Metrics

- **Enhanced Time-series Aware Precision & Recall, eTaPR**
  - Imprecise of insufficient detections are not suitable in expert scenarios
  - Cross referencing prevents scoring inaccurate predictions or insufficiently detected anomalies
    - $A^d = \left\{ a \mid a \in A \text{ and } \frac{\sum_{p \in P^c} |a \cap p|}{|a|} \geq \theta_r \right\}$
    - $P^c = \left\{ p \mid p \in P \text{ and } \frac{\sum_{a \in A^d} |a \cap p|}{|p|} \geq \theta_p \right\}$
    - P $\rightarrow \boldsymbol{A^d}(1)$, $\boldsymbol{A^d(1)} \rightarrow \boldsymbol{P^c(1)}$
    - $\boldsymbol{P^c(1)} \rightarrow \boldsymbol{A^d(2)}$, $\boldsymbol{A^d(2)} \rightarrow \boldsymbol{P^c(2)}$, $\boldsymbol{P^c(2)} \rightarrow \boldsymbol{A^d(3)}$ …

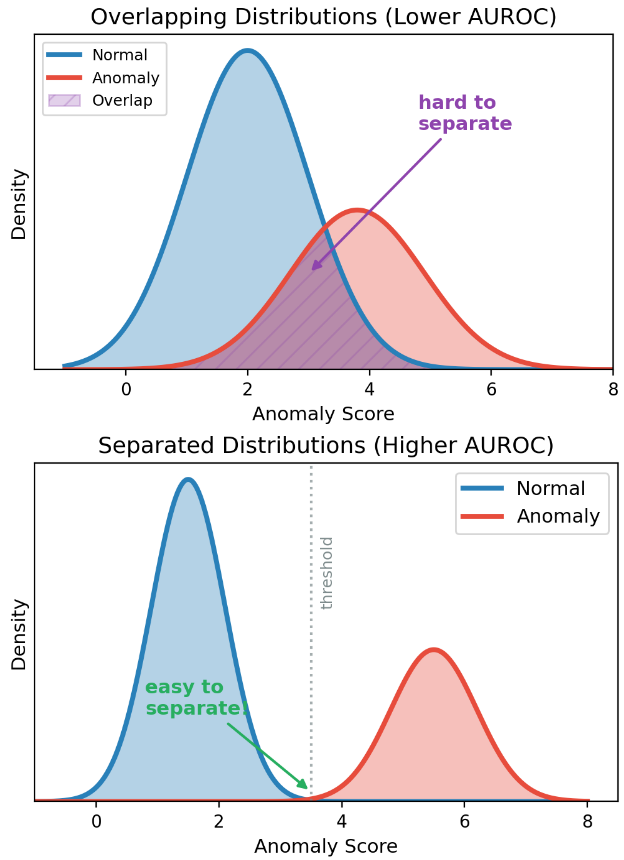
    until $\boldsymbol{P^c(i)} = \boldsymbol{P^c(i-1)}$ and $\boldsymbol{A^d(i)} = \boldsymbol{A^d(i-1)}$
  - A length-based weighting scheme reflects prediction accuracy in precision-like score

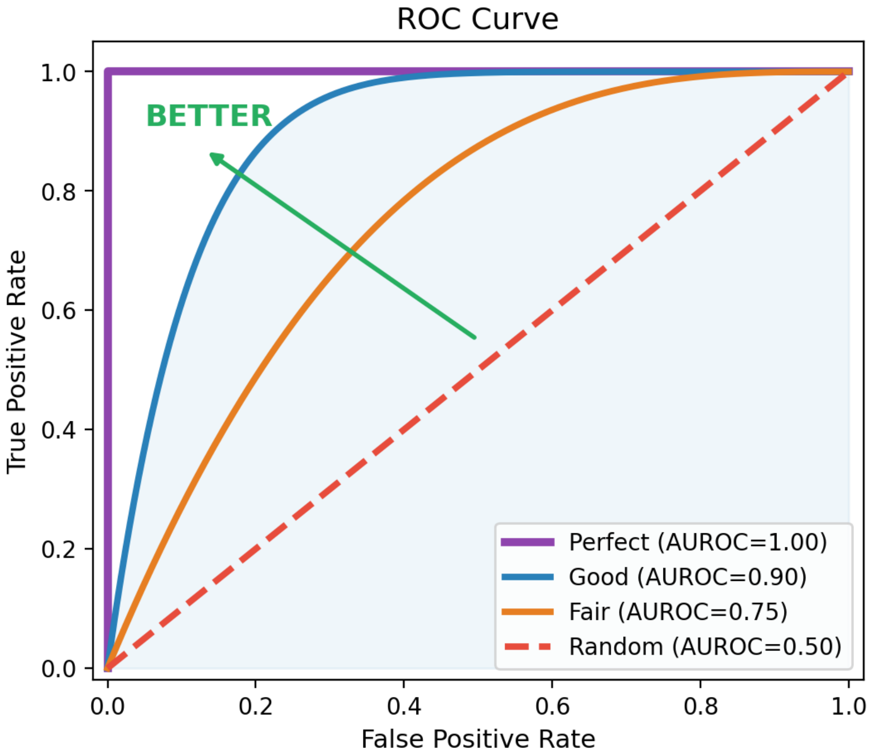




Multivariate Time-Series Anomaly Detection-Evaluation Metrics

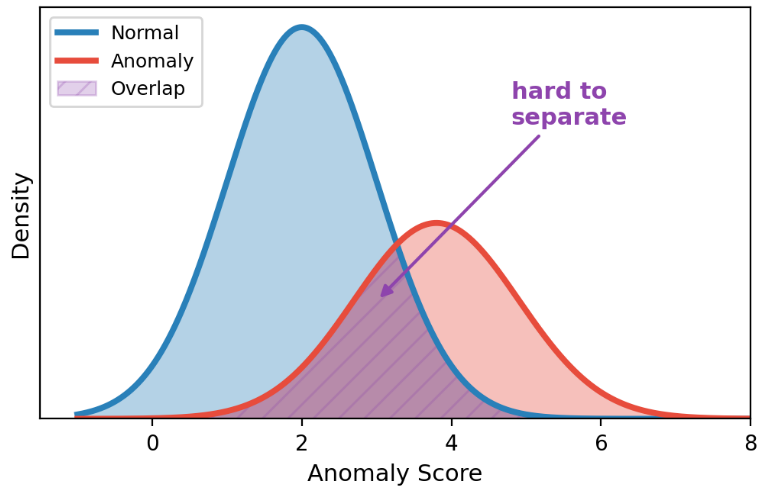
- **AUROC**
  - Measures how well a model separates normal and anomalous samples
  - Computed from the ROC curve, which plots **True Positive Rate vs False Positive Rate** across different thresholds
  - Threshold-independent evaluation metric

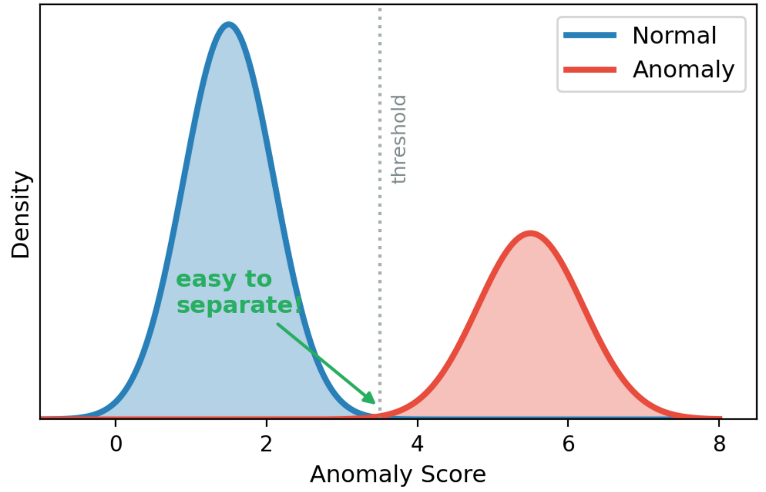




Multivariate Time-Series Anomaly Detection-Threshold

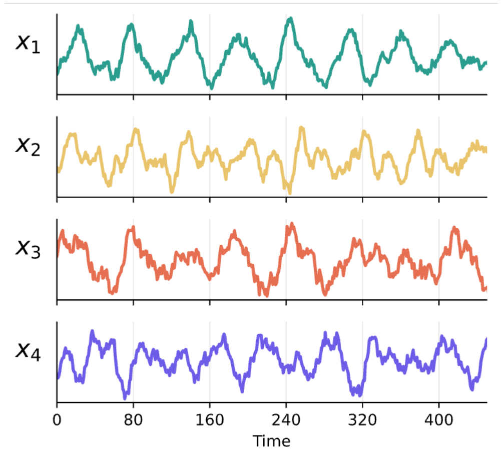
- **Gaussian Assumption-based 3-sigma**
  - Estimate the normal score distribution as Gaussian and set the threshold at three standard deviations above the mean
- **Percentile-based Threshold**
  - Set the threshold at the upper percentile of the normal score distribution
- **Maximum Score on the Validation Set**
  - Use the maximum anomaly score observed on the validation set as the threshold
- **POT based on Extreme Value Theory**
  - Tail distribution of extremes converging to a fixed form regardless of overall data
  - Mathematical thresholding for anomaly detection using this principle
- **Best F1 Search**
  - Evaluate F1 scores over multiple candidate thresholds and choose the threshold that gives the best result

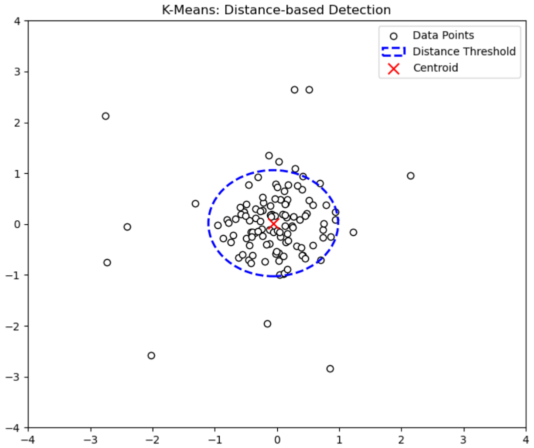


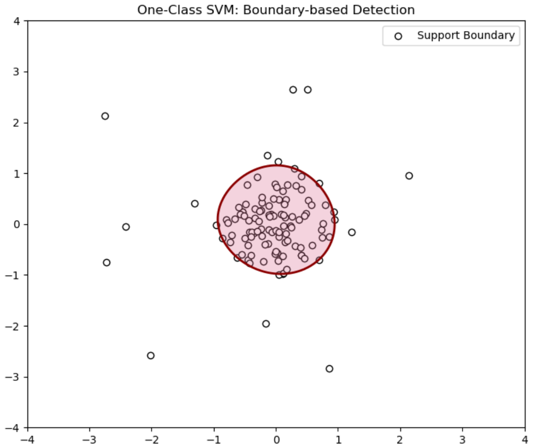


Multivariate Time-Series Anomaly Detection-Previous Work

- **Traditional Machine Learning-based Approaches**
  - Core assumption: anomalies are different from normal samples
- **Examples**
  - k-means / clustering-based methods
  - kNN
  - OCSVM
- **Characteristics**
  - Focus on distance- or boundary-based discrimination rather than strongly modeling the intrinsic temporal structure of time series
  - Often treat each time point or window as a data point in feature space

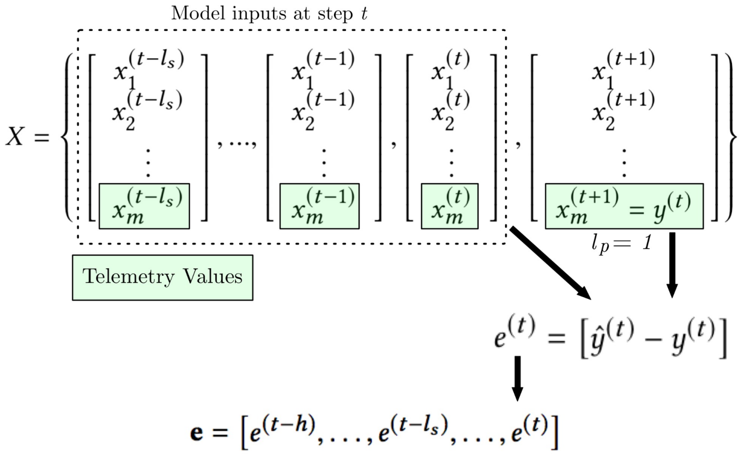


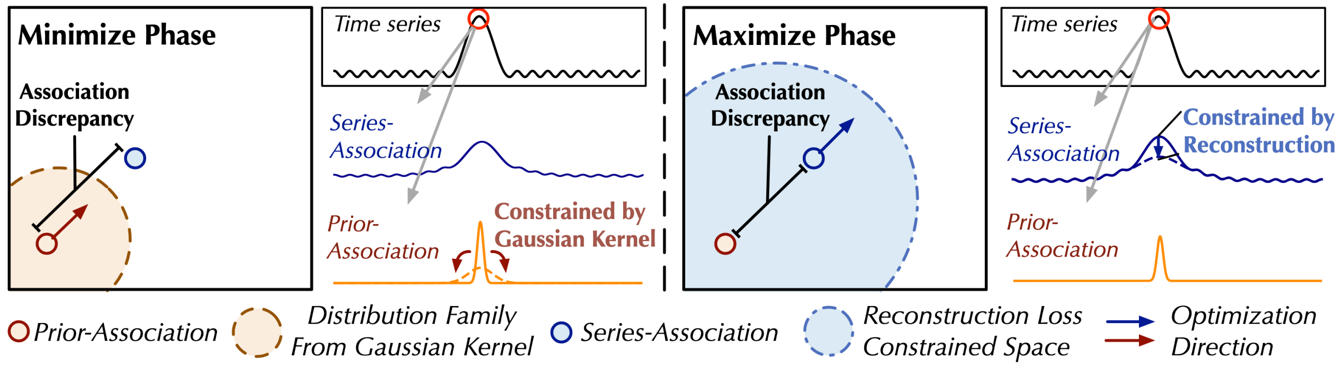




Multivariate Time-Series Anomaly Detection-Previous Work

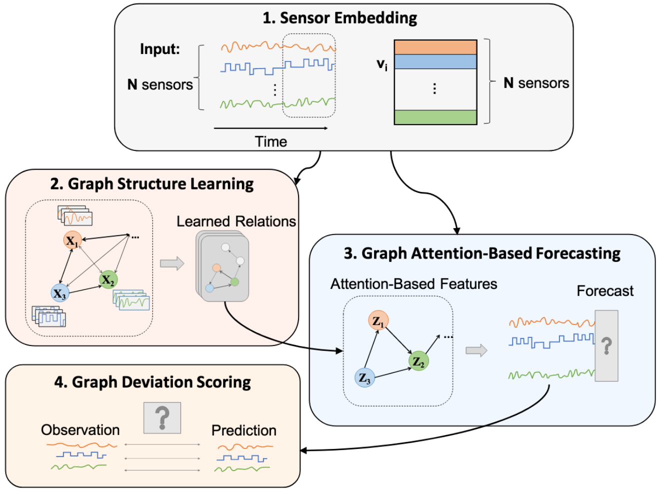
- **Approaches Focused on Temporal Patterns in Time-Series**
  - LSTM-NDT
    - Learns temporal patterns for each individual series using an LSTM
  - Anomaly Transformer
    - Assumes that anomaly points are mainly associated only with very nearby time points and have weak overall connectivity within the window
    - Assumes that normal points have not only local dependencies but also global relations across the entire series
    - Detects anomalies based on the discrepancy between prior association (a learnable Gaussian for adjacent relations) and series association (self-attention across time points within the input window)
    - A larger discrepancy indicates normality, while a smaller discrepancy indicates anomaly

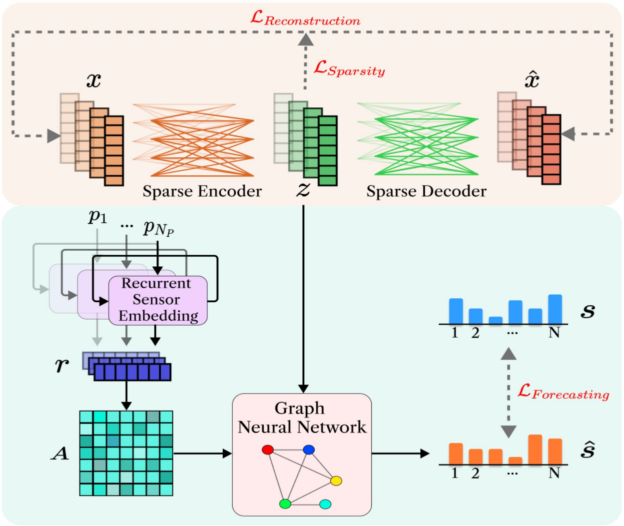




Multivariate Time-Series Anomaly Detection-Previous Work

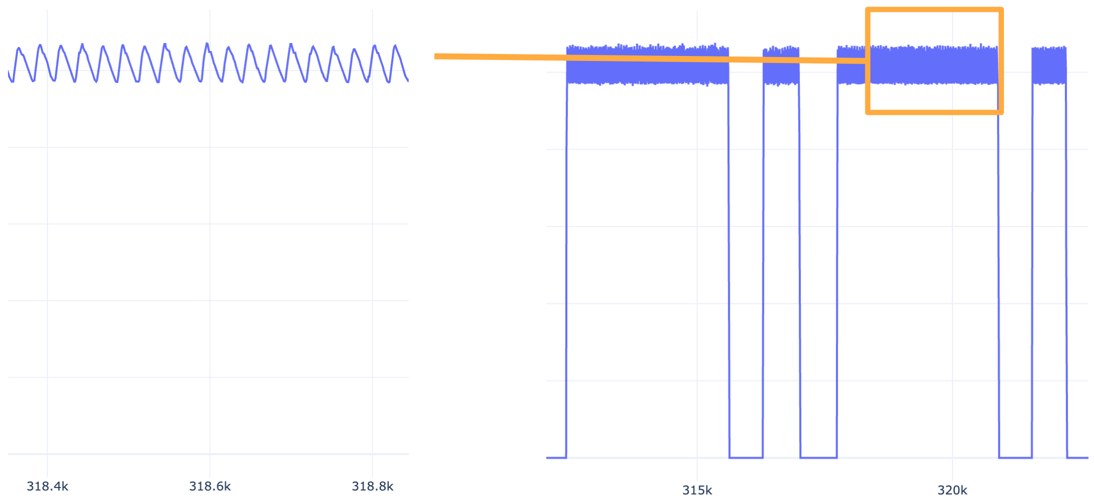
- **Approaches that Model Inter-Series Relationships**
  - Anomaly Transformer also captured inter-series relationships implicitly
  - GDN
    - Constructs a graph based on series profiles
    - Generates node embeddings from both series profiles and the input window
  - FuSAGNet
    - Follows a similar direction as GDN, but builds process-wise series profiles
    - Transforms the input window through a sparse encoder-decoder and combines it with series profiles to generate node embeddings





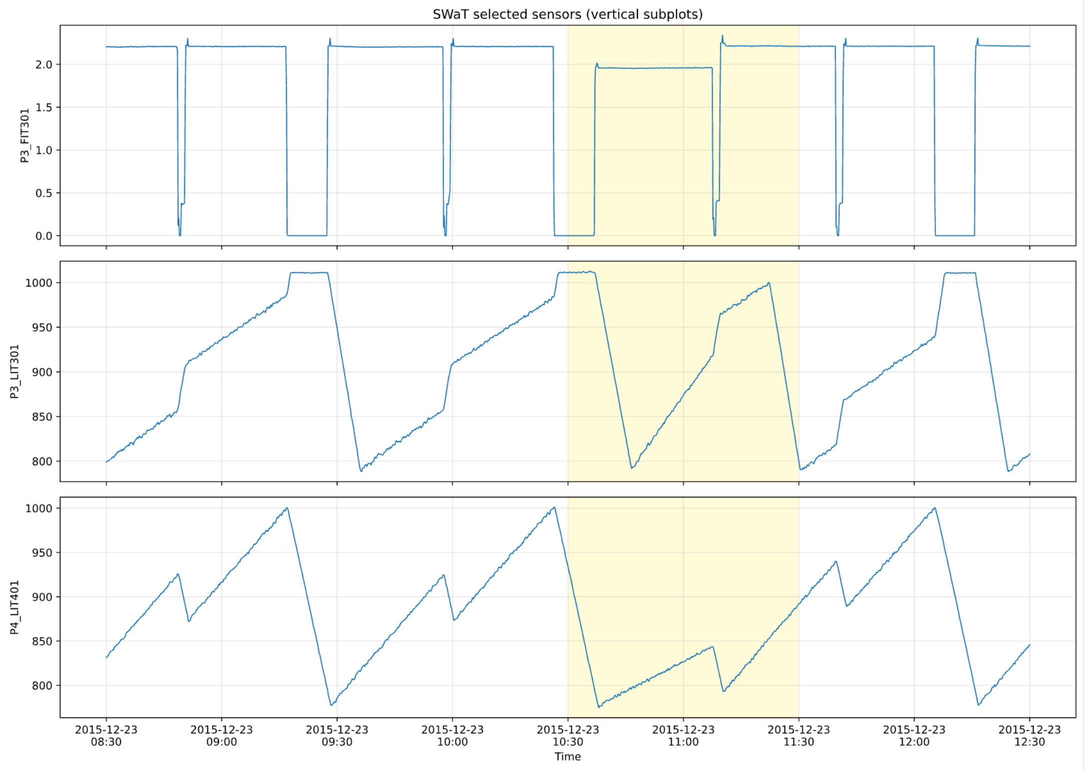
Motivation and Key Observations

- **Temporal Perspective**
  - Time series are sequential data shaped by past observations
  - Their values reflect both short-term fluctuations and long-term trends
    - Raw signals capture fine-grained local changes
    - Long-term trends capture overall dynamics and inertia
  - Focusing only on local variations may overreact to noise
  - Focusing only on long-term trends may miss important local patterns



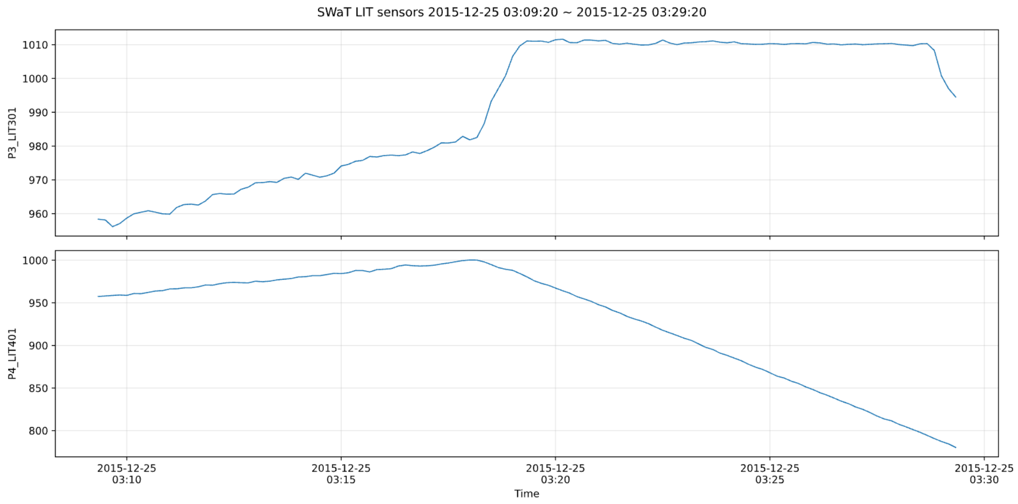
Motivation and Key Observations

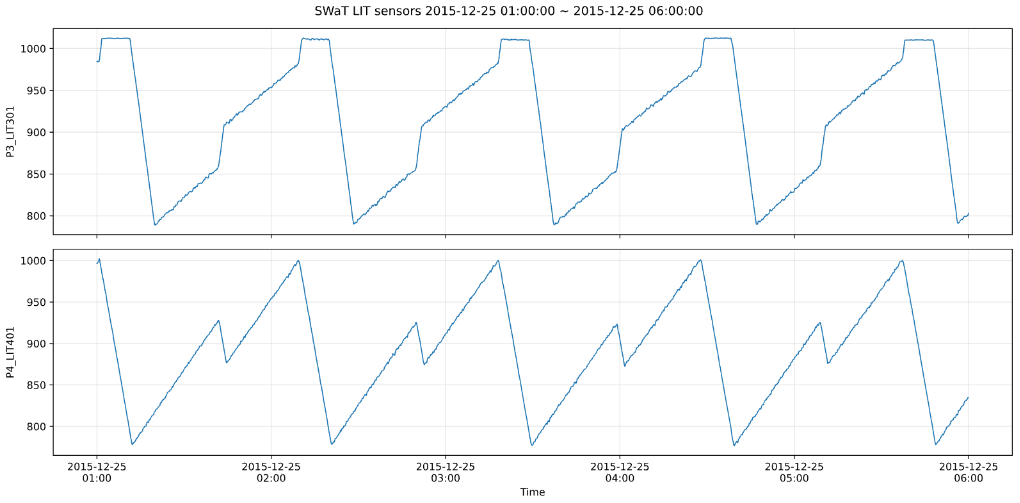
- **Inter-Series Relational Perspective**
  - Series may evolve independently or interact with each other
  - Inter-series relationships can be complex and non-linear
  - Anomalies may appear as broken interactions across series
  - Modeling each series independently may miss relation-based anomalies



Motivation and Key Observations

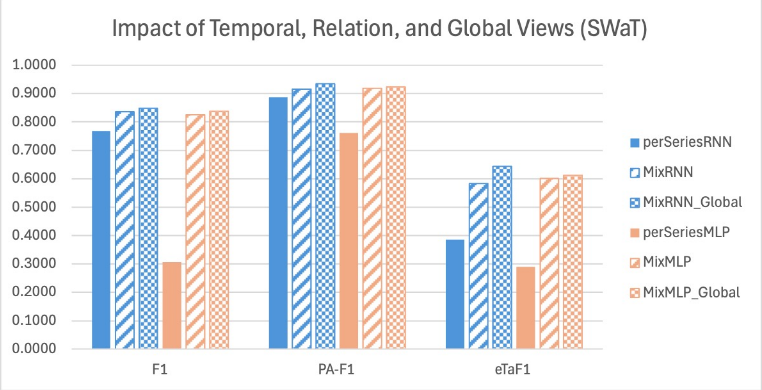
- **Global Contextual Perspective**
  - Input windows are limited by efficiency and real-time constraints
  - Series that look different locally may behave similarly at a larger scale
  - Similar-looking series may still require different interpretations
  - Global context is needed for series-specific interpretation

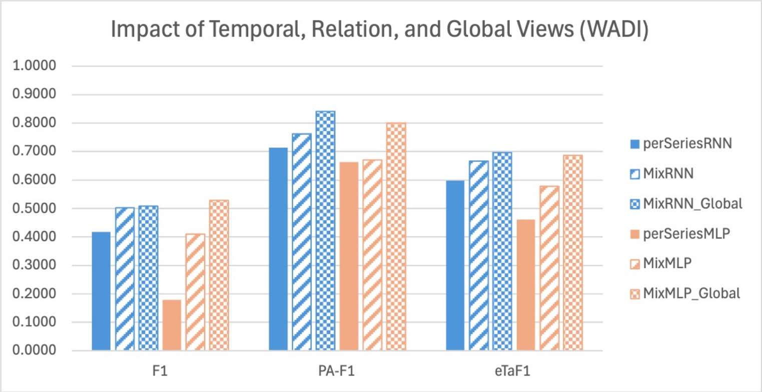




Motivation and Key Observations

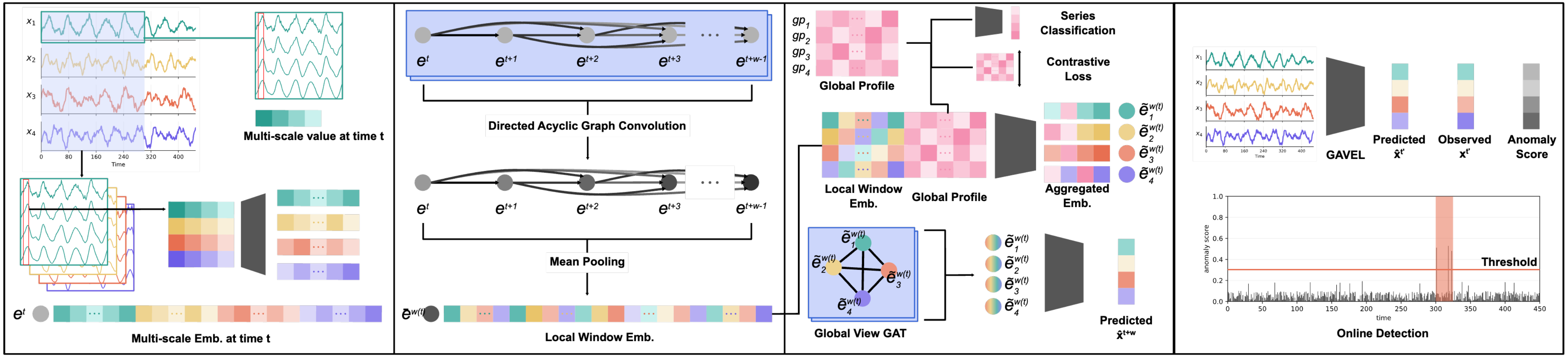
- **Preliminary Experiments**
  - perSeries: temporal perspective
  - mix: temporal + relational perspective
  - global: interpreting relationships under global context
  - Performance improved consistently as more perspectives were incorporated
  - The benefit of adding global context was more pronounced on WADI than on SWaT





	SWaT			WADI		
Model	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
perSeriesRNN	0.7688	0.8874	0.3858	0.4181	0.7132	0.5983
MixRNN	0.8364	0.9150	0.5835	0.5035	0.7615	0.6672
MixRNN_Global	**0.8485**	**0.9347**	**0.6437**	**0.5087**	**0.8415**	**0.6963**
perSeriesMLP	0.3060	0.7617	0.2898	0.1791	0.6631	0.4611
MixMLP	0.8250	0.9180	0.6006	0.4098	0.6705	0.5780
MixMLP_Global	0.8377	0.9243	0.6119	0.5286	0.8009	0.6865

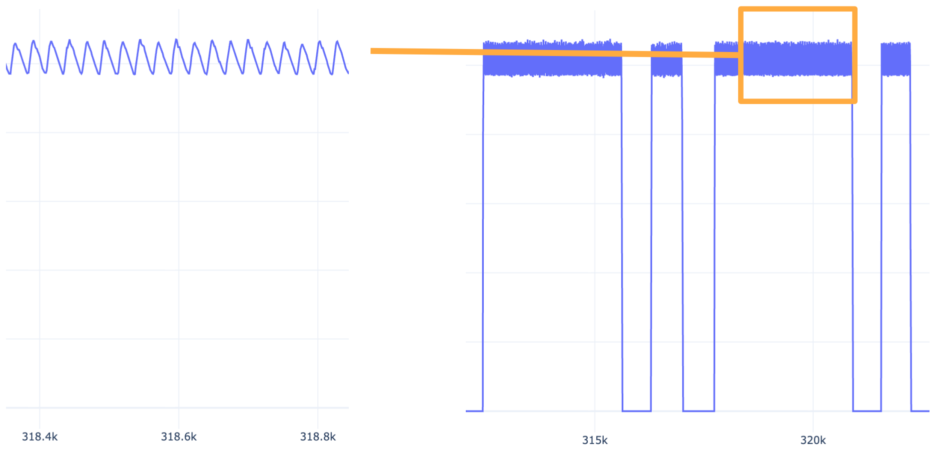
- **Overview**

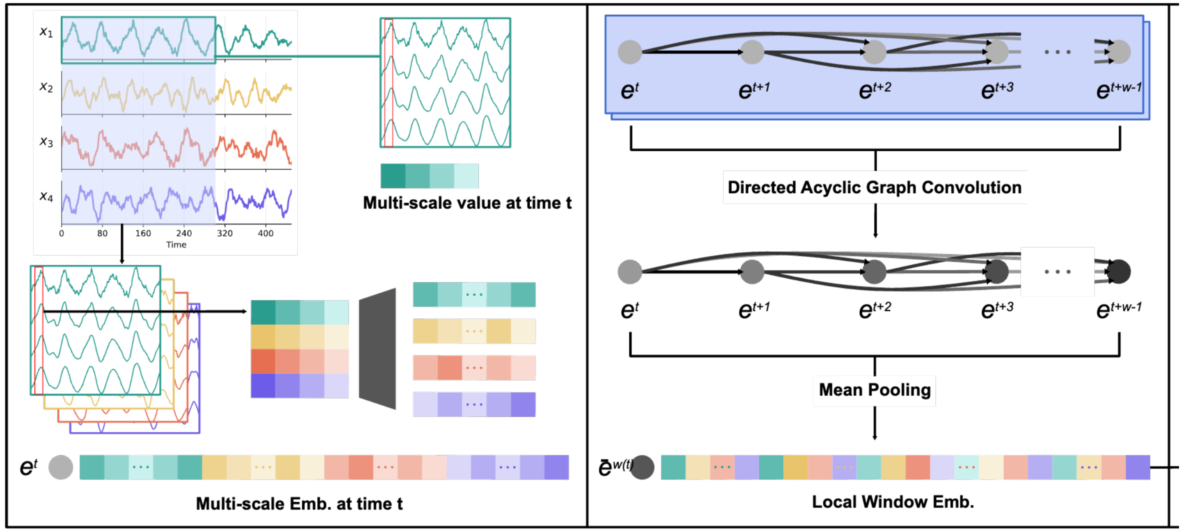


Multi-scale value at time t
Multi-scale Emb. at time t
Directed Acyclic Graph Convolution
Mean Pooling
Local Window Emb.
Global Profile
Series Classification
Contrastive Loss
Local Window Emb.
Global Profile
Aggregated Emb.
Global View GAT
Predicted
GAVEL
Predicted
Observed
Anomaly Score
Threshold
Online Detection

GAVEL

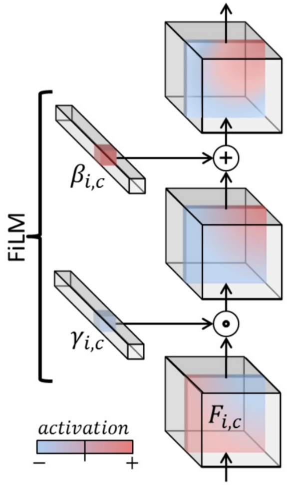
- **Local Window Encoder**
  - Directed Acyclic Graph
    - Encodes the window from a temporal perspective
    - Models local dependencies across time steps
    - Treats each time step as a node with directed temporal edges
    - Captures both temporal and inter-series relational patterns within the window
  - Multi-Scale Node Embedding
    - Single-scale representations may be noise-sensitive or miss long-term trends
    - Combines short-term and long-term temporal patterns
    - Builds node embeddings from the raw signal and multi-scale moving averages

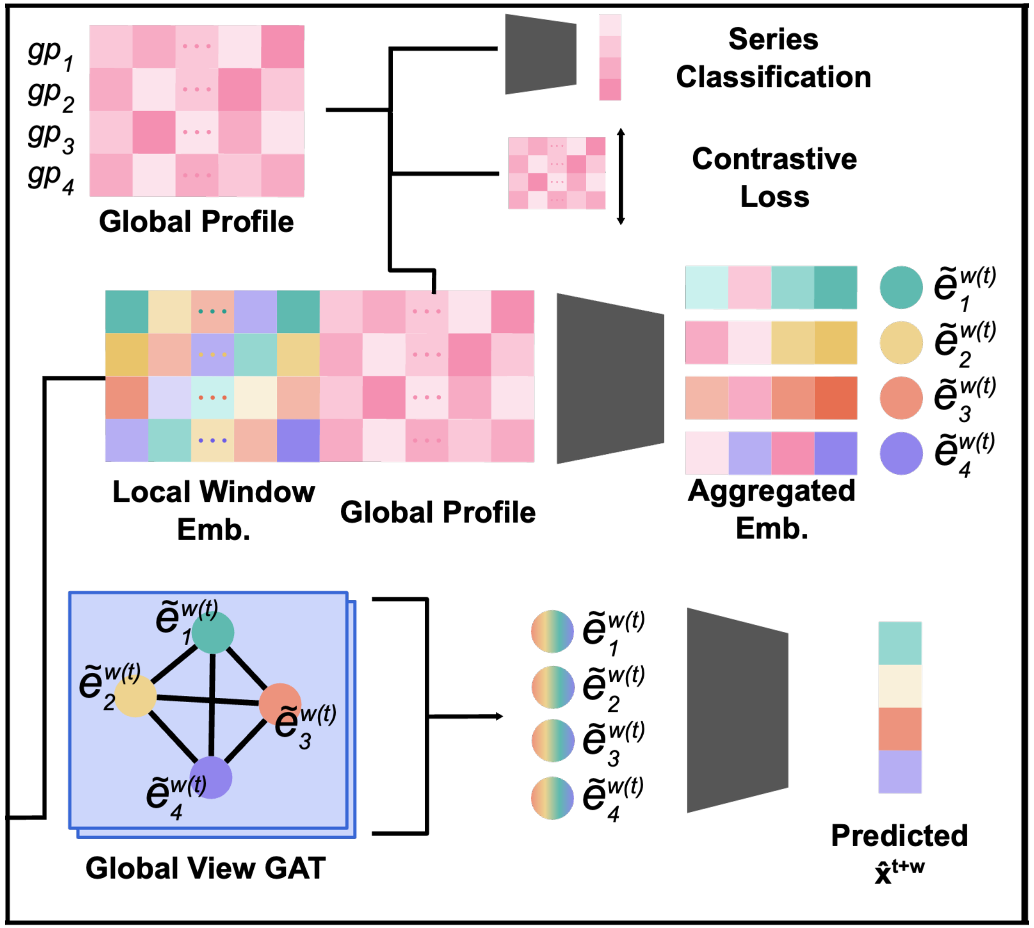




GAVEL

- **Global View Decoder**
  - Global Profile
    - Combines each window representation with a global profile
    - Uses contrastive learning to make series profiles more discriminative
  - FiLM
    - Fuses local embeddings with global profiles
    - Injects series-specific characteristics through feature-wise modulation
  - GAT
    - Treats series-wise representations as nodes
    - Learns inter-series interactions with graph attention





RQ1 Main Table

- **Motivation**
  - Check whether the proposed method outperforms the baselines
- **Design**
  - Include TSMixer as a relevant baseline for temporal and relational modeling (TSMixer was originally proposed for forecasting)
  - Use the best reported hyperparameters and window size for each method
  - Standardize preprocessing, batch size, learning rate, optimizer, and thresholding

Method	**Intra-series**	**Inter-series**	**Global**
kNN	X	X	X
OCSVM	X	X	X
AT (ICLR '22)	O	O	X
USAD (KDD '20)	O	O	X
TSMixer (TMLR '23)	O	O	X
FuSAGNet (KDD '22)	O	O	O
Ours	O	O	O

	SWaT			WADI		
Model	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
KNN	0.7831	0.8461	0.3812	0.2263	0.2710	0.2759
OCSVM	0.7809	0.8461	0.3576	0.1153	0.1489	0.1495
DeepSVDD (ICML '18)	0.7300	0.7961	0.0917	0.0164	0.0319	0.0319
USAD (KDD '20)	0.8250	0.8884	0.4831	0.4101	0.8386	0.5990
TSMixer (TMLR '23)	0.7836	0.9085	0.5878	0.2829	0.7058	0.5063
AnomalyTrans (ICLR '22)	0.7881	0.8745	0.4280	0.2912	0.6052	0.4166
FuSAGNet (KDD '22)	0.8469	0.9302	0.6319	0.5414	0.7891	0.6733
GAVEL	**0.8729**	**0.9431**	**0.7145**	**0.7078**	**0.9342**	**0.7754**

RQ2 Local Window Encoder

- **Motivation**
  - Validate whether key design choices inside the Local Window Encoder translate into measurable performance gains
- **Design**
  - Series-Mix (Local Relation)
    - Use group conv1d to assign independent per-series weights while blocking cross-series information exchange
  - DAG Edge Topology
    - Replace DAG edges with a fully connected temporal graph so all time nodes can interact
  - Multi-Scale Representation
    - Remove multi-scale embedding and use a single-scale embedding projection from raw values only
- **Results and Interpretation**
  - Removing any single component degrades performance, indicating all three components contribute to the local encoder

	Ablation			SWaT			WADI		
Model	Series-Mix	DAG edge	Multi-Scale	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
	X	O	O	0.8589	0.9260	0.6299	0.4959	0.8400	0.6698
	O	X	O	0.8626	0.9311	0.6488	0.5767	0.8380	0.7114
	O	O	X	0.8649	0.9365	0.6864	0.6173	0.9050	0.7452
GAVEL	O	O	O	**0.8729**	**0.9431**	**0.7145**	**0.7078**	**0.9342**	**0.7754**

RQ3 Global View Decoder

- **Motivation**
  - Validate the effect of global-view injection in the Global View Decoder
- **Design**
  - No-Global / No-FiLM baseline
    - Remove both the global profile and FiLM to form a strict baseline
  - Series-wise vs Process-wise vs Shared Profile (FiLM)
    - Compare series-wise, process-wise, and shared profiles under the same FiLM setting
  - FiLM vs Concat
    - Compare FiLM and Concat under matched settings, with Concat experiments

	Ablation		SWaT			WADI		
Model	Cond. Inject.	Cond. Method	F1	PA-F1	eTaF1	F1	PA-F1	eTaF1
GAVEL	X	X	0.8153	0.8850	0.5510	0.3523	0.4263	0.4574
GAVEL	Concat	Shared	0.8519	0.9252	0.6469	0.6372	0.9280	0.7590
GAVEL	Concat	Process-wise	0.8493	0.9278	0.6398	0.6156	0.8855	0.7488
GAVEL	Concat	Series-wise	0.8612	0.9285	0.6644	0.6016	0.9077	0.7385
GAVEL	FiLM	Shared	0.8694	0.9367	0.6654	0.5871	0.8856	0.7447
GAVEL	FiLM	Process-wise	0.8651	0.9437	0.6619	0.6225	0.9162	0.7442
GAVEL	FiLM	Series-wise	**0.8729**	**0.9431**	**0.7145**	**0.7078**	**0.9342**	**0.7754**

RQ4 Auxiliary Loss

- **Motivation**
  - Effect of Euclidean distance loss and classification loss on global profile quality
- **Design**
  - Weights tested from 0.0 to 0.4 in 0.1 increments

SWaT	cls \ euc	0	0.1	0.2	0.3	0.4
F1	0	0.8636	0.8720	0.8604	0.8629	0.8728
	0.1	0.8584	0.8644	0.8727	**0.8729**	0.8685
	0.2	0.8626	0.8410	0.8517	0.8558	0.8688
	0.3	0.8650	0.8579	0.8563	0.8541	0.8464
	0.4	0.8638	0.8727	0.8635	0.8628	0.8539

SWaT	cls \ euc	0	0.1	0.2	0.3	0.4
PA-F1	0	0.9316	0.9372	0.9316	0.9421	0.9485
	0.1	0.9286	0.9310	0.9422	**0.9431**	0.9370
	0.2	0.9381	0.9130	0.9307	0.9328	0.9351
	0.3	0.9372	0.9212	0.9281	0.9371	0.9366
	0.4	0.9342	0.9463	0.9383	0.9302	0.9337

SWaT	cls \ euc	0	0.1	0.2	0.3	0.4
eTaF1	0	0.6560	0.6739	0.6539	0.6771	0.6946
	0.1	0.6194	0.6407	0.6862	**0.7145**	0.6944
	0.2	0.6865	0.5858	0.6335	0.6531	0.6799
	0.3	0.6815	0.6243	0.6186	0.6615	0.6331
	0.4	0.6728	0.7016	0.6672	0.6606	0.6570

WADI	cls \ euc	0	0.1	0.2	0.3	0.4
F1	0	0.5880	0.7078	0.6451	0.7517	0.6217
	0.1	0.6341	0.6007	0.6295	0.6136	0.6521
	0.2	0.6864	0.6786	0.5788	0.6573	0.6773
	0.3	0.5852	0.7175	**0.7204**	0.6500	0.6215
	0.4	0.6585	0.6244	0.6590	0.6751	0.6487

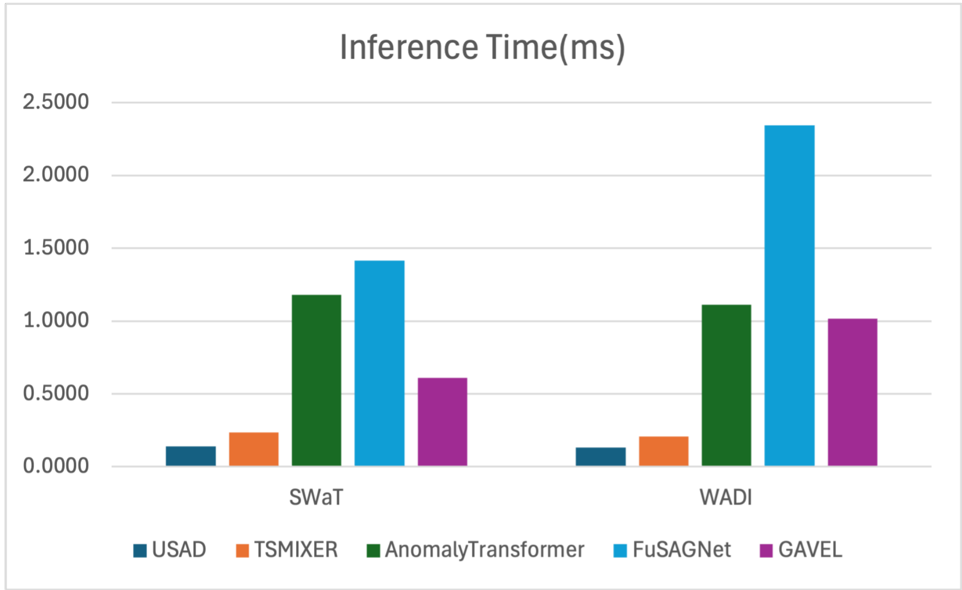
WADI	cls \ euc	0	0.1	0.2	0.3	0.4
PA-F1	0	0.9299	0.9342	0.8986	0.9300	0.8409
	0.1	0.9489	0.9007	0.9111	0.9318	0.8647
	0.2	0.9639	0.9182	0.8957	0.9319	0.8741
	0.3	0.8926	0.9412	**0.9537**	0.8765	0.8731
	0.4	0.8706	0.9200	0.8841	0.8856	0.8713

WADI	cls \ euc	0	0.1	0.2	0.3	0.4
eTaF1	0	0.7364	0.7754	0.7606	0.7878	0.7268
	0.1	0.7663	0.7344	0.7591	0.7473	0.7326
	0.2	0.7946	0.7534	0.7109	0.7564	0.7602
	0.3	0.7235	0.7833	**0.7954**	0.7125	0.7365
	0.4	0.7588	0.7539	0.7444	0.7615	0.7371

RQ5 Inference Time

- **Motivation**
  - Real-time feasibility for online detection
- **Design**
  - Inference time per window (ms), batch size 1, RTX 4080 SUPER
  - Average of 5 runs after 2 warm-up runs

	Inference Time(ms)	
model_name	SWaT	WADI
USAD	0.1385	0.1335
TSMIXER	0.2369	0.2095
AnomalyTransformer	1.1795	1.1116
FuSAGNet	1.4151	2.3414
GAVEL	0.6081	1.0172



Conclusion

- **Capturing Temporal Dynamics Together with Local/Global Relations Is Crucial for TSAD**
- **We Propose GAVEL, Which Integrates a Local Window Encoder and a Global View Decoder**
- **GAVEL Achieves Consistent Gains across Multiple Metrics, with Ablations Validating Each Component**